Vibrational stabilization by an internal binary: a negative result

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We test whether a tight central binary can stabilize, by analogy with the Kapitza pendulum, an equilateral triangle of three equal masses that is unstable with a single central mass. We compare the survival time with that of a control group with the same total mass concentrated in a single body, for three central masses, ten binary separations, and both senses of rotation, and we repeat the test at the Routh stability boundary of the Lagrange triangle. In no case did the binary significantly prolong survival; for larger separations, it shortened survival to about 8% of the control. A scaling estimate explains the result: since Kepler's third law links the amplitude and frequency of the quadrupole excitation, the effective Kapitza term decays with the seventh power of the binary separation. Passive vibrational stabilization does not occur in this system.

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I. INTRODUCTION

An inverted pendulum can be stabilized by rapid vibration of its suspension point, an effect discovered by Kapitza [1]. In the high-frequency limit, the averaged effect of the excitation enters the slow dynamics as an effective potential proportional to the square of the force amplitude and inversely proportional to the square of the frequency [2]. This result suggests a natural question in celestial mechanics: could a tight internal binary, whose gravitational field oscillates rapidly, stabilize a three-body configuration that is unstable without it?

This note describes a numerical test of this hypothesis, with a control group, and a physical constraint that explains the negative result obtained. The hypothesis was chosen because it is small (a specific case), testable (the simulation can refute it), and nontrivial (a positive result would be surprising).

II. METHOD

A. Configurations and groups

The reference configuration is an equilateral triangle of three equal masses ($m = 1$) at distance $R = 1$ from a central body, rotating rigidly. With equal masses, this configuration is unstable, with or without a central body.

- **Test group:** at the center, a circular binary of total mass M, equal components, and separation a_b, rotating in the same sense as the triangle or in the opposite sense.
- **Control group:** at the center, a single mass M, which reproduces the monopole term of the binary without the oscillating component.
- **Perturbation:** displacements of 10^{-4} in position or velocity, identical in both groups; median of three perturbations.

The initial orbital velocity of the triangle includes the mean quadrupole correction of the binary,

$$v^2 = \frac{1}{R}\left[GM\left(1+\frac{3}{16}\frac{a_b^2}{R^2}\right)+\frac{Gm}{\sqrt{3}}\right], \tag{1}$$

so that the test group does not start out of equilibrium by construction.

B. Integration and disruption criterion

The integration uses the symplectic kick-drift-kick leapfrog method, with a step of $P_b/60$, where P_b is the binary period, capped at 0.002 time units, in units with $G = 1$. The triangle is considered disrupted when the representative point on the shape sphere [7] falls below 53° latitude; the pole corresponds to the equilateral configuration.

III. RESULTS

A. Triangle with a central binary

Figure 1 shows the ratio of the survival time with the binary to that of the control, for three central masses. Values above 1 would indicate stabilization.

In no combination did the binary significantly prolong survival. For $a_b \lesssim 0.07$, the result coincides with the control; for larger separations, the binary shortens the lifetime of the triangle, down to about 8% of the control survival time for $M = 10$ and $a_b = 0.6$. Reversing the sense of rotation of the binary did not change the pattern.

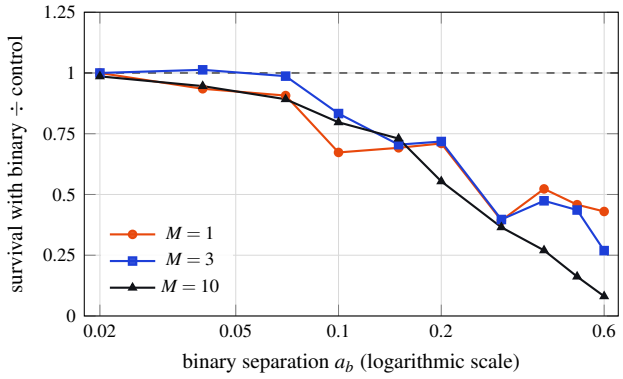


Fig. 1. Median survival time of the triangle with a central binary, divided by that of the single-mass control, as a function of the binary separation, for three central masses. The dashed line indicates equality with the control. No point exceeds it significantly.

B. Binary at the Routh boundary

A second test placed the binary in the position of the dominant body of a Lagrange triangle with masses $1 : \mu : \mu$, in the vicinity of the Routh stability limit ($\mu \approx 0.0198$) [8], where small effects could tip the outcome. Table I summarizes the median survival times.

The increase observed at $\mu = 0.022$ for $a_b = 0.01$ and 0.02 occurs exactly on the boundary, where survival varies greatly between perturbations, and is not repeated at $\mu = 0.025$. There is no evidence of stabilization.

IV. DISCUSSION

A. The Keplerian constraint

In the Kapitza pendulum, the amplitude and frequency of the excitation are independent. In a gravitational binary, they are not: Kepler's third law links them. Let r be the distance from an external body to the binary. The dominant oscillating component of the acceleration is quadrupolar, with amplitude

$$F \sim \frac{GMa_b^2}{r^4}, \tag{2}$$

and frequency of order $\omega_b = \sqrt{GM/a_b^3}$. In the Kapitza regime, the averaged effect of the rapid excitation enters as an effective potential proportional to F^2/ω^2 [2]. Combining the two relations,

$$\frac{F^2}{\omega_b^2} \sim \frac{GM}{r^8}\, a_b^7. \tag{3}$$

The stabilizing term decays with the seventh power of the separation. A tight binary oscillates rapidly, a necessary condition for the averaging regime, but produces a negligible effective term. A wide binary produces strong but slow excitation and begins to resonate with the eigenfrequencies of the triangle, which destabilizes it. The estimate is a scaling argument and does not replace a complete analysis, but it is consistent with the two observed regimes.

B. Relation to the literature

Stabilization of chaotic solutions in the restricted problem has been achieved with additional bodies placed at the triangular Lagrange points [3], and unstable periodic orbits of the three-body problem have been stabilized with small impulses, one per period, computed from the linearized Poincaré map [4], in the spirit of the chaos control of Ott, Grebogi, and Yorke [6]. In both cases, stabilization arises from positioned bodies or from active control, not from passive gravitational excitation. For circumbinary planets, averaged models of the quadrupole potential describe the secular dynamics well, but do not capture the resonances between the binary period and that of the outer orbit, which dominate the instability near the binary [5].

C. Limitations

- Planar simulation, with a single family of initial configurations.
- Disruption criterion defined by the experiment (latitude of 53° on the shape sphere).
- Three perturbations per point; results at the stability boundary would require larger samples.
- Inclined orbits and eccentric binaries, which alter the excitation spectrum, were not explored.

V. CONCLUSIONS

With the caveats above, the result is negative and has a simple physical explanation: gravity does not allow the amplitude and frequency of the vibration to be chosen freely, and the combination permitted by Kepler's law does not reach the Kapitza regime. A tight binary is, for practical purposes, equivalent to a single mass; a wide binary destabilizes through resonance.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

Table I. Median survival time, in time units, of the Lagrange triangle $1:\mu:\mu$ with the dominant body replaced by a binary of separation a_b. Integration limit: 1500 time units.

μ	Control	$a_b = 0.005$	0.01	0.02	0.04	0.1	0.2
0.015	stable	stable	stable	stable	stable	stable	stable
0.022	150	135	205	241	62	46	28
0.025	42	42	42	42	42	35	23
0.030	32	32	32	32	32	26	18

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

The author declares no conflicts of interest. This work received no funding. This text has not undergone peer review and should be read as a working note. Corrections are recorded in the note itself, with the date.

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